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Atomic layer bonding contacts in two-dimensional semiconductors

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Van der Waals contact between two-dimensional semiconductors and metals has always been inferior to covalent bond contacts used in semiconductor industry because of weak band coupling and low bond strength. Here, we report an atomic layer bonding (ALB) contact with strong band coupling and high interfacial cohesion by establishing a metallic coherent bonding interface between the transition-metal atomic layer of transition-metal dichalcogenides and metals. This contact exhibits ultralow contact resistance and superb thermomechanical stability, comparable to those of covalent bond contacts and surpassing all reported contact configurations. ALB contact formed in monolayer molybdenum disulfide and gold demonstrates a contact resistance of 70 ohm-micrometers and thermomechanical stability up to 400°C and delivers a maximum on-current of 1.1 milliamperes per micrometer after high-temperature annealing, all of which meet industrial integration.

Metal-semiconductor contacts are crucial for the development of high-performance electronic devices and their industrial integration, especially as device sizes continue to shrink. Industrial metal-semiconductor contacts necessitate ultralow contact resistance (R_C), superior thermomechanical stability up to 400°C for compatibility with back-end-of-line (BEOL) processes, enhanced thermal and electrical conductivity for heat dissipation, and seamless integration with standard semiconductor techniques. Covalently bonded metal-semiconductor contacts, exemplified by silicon-silicide (Si-M_xSi_y) (*1*), titanium nitride–gallium nitride (TiN-GaN) (*2*), and titanium silicon carbide–silicon carbide (Ti_3SiC_2-SiC) (*3*), represent leading strategies in the semiconductor industry. These contacts exhibit strong interfacial band coupling and high bonding strength, resulting in ultralow R_C and exceptional thermomechanical stability, both of which are most critical for industrial integration (*2*, *4*). However, achieving strong bonding contact in two-dimensional (2D) van der Waals (vdW) semiconductors such as transition-metal dichalcogenides (TMDs) is highly challenging because of the vdW gap and dangling-bond-free surface. The vdW gap blocks charge transfer between the metal and semiconductor, and the dangling-bond-free surface drastically reduces the likelihood of chemical bonding. As a result, the current contact between 2D TMDs and metals is dominated by vdW contacts. The inherent weak band coupling and low bonding strength of the vdW contacts inevitably lead to two limitations: large tunnel barrier [i.e., high R_C (*5*–*9*)] and low contact stability.

Attempts to overcome the limitations of vdW contact include edge contacts (*10*, *11*), low–work function metals (*12*–*14*), ultrahigh vacuum evaporation (*15*, *16*), ultrashallow Y-doping contact (*17*), and the recently much-anticipated semimetal contacts (*18*–*23*). These efforts have reduced the R_C to tens of ohm-micrometers. However, almost all existing contact schemes have failed to break the weak bonding of vdW interfaces. R_C and thermomechanical stability, especially up to 400°C required for BEOL integration, have not yet fully synchronized and are far inferior to strong covalent bonding interfaces, hindering the development of "lab to fab." A prime example is that the vdW contacts between semimetal bismuth or antimony (Bi/Sb) and 2D TMDs either cannot meet the thermal budget for BEOL integration (fig. S1A) or the R_C has increased substantially after heating at <400°C (*14*, *18*, *19*, *24*). In addition, thermal deformation of electrodes at weakly bonded interface threatens contact stability (*25*, *26*).

Here, we report an atomic layer bonding (ALB) contact (Fig. 1A) for 2D TMD semiconductors and metals. We established a metallic coherent bonding interface between the transition-metal atomic layer and the metal electrode by precisely removing the upper sulfur-group atoms, matching electronic structures and lattice structure. ALB contact exhibits strong band coupling and high interfacial cohesion (Fig. 1, C and E), comprehensively overcoming the limitations of vdW contact (Fig. 1, D and E). As a result, we achieved a breakthrough in both ultralow R_C and excellent thermomechanical stability, comparable to those of covalent bonding contacts in the semiconductor industry.

Comparison of ALB contact and vdW contact

Realizing the ALB contact between transition-metal atomic layer in TMDs and metal is theoretically an ideal way to enhance bonding and achieve overall breakthrough in contact performance. This is mainly attributed to the multidirectional and far-extending properties of the d orbitals in transition-metal atoms, which have high bonding probability and strong bonding ability with metal electrodes. That is, the electron on the d orbital of transition-metal atoms can form strong metallic bonds with the electron of the metal at the interface. This provides a favorable electronic structure foundation. Through density functional theory (DFT) calculations, we found that after removing the upper S atomic layer, the Mo atomic layer in monolayer MoS_2 can coherently bond with the Au (111) atoms through lattice relaxation, and the charge transfer and accumulation at the bonding interface are markedly higher than those at the vdW interface (Fig. 1, A and B). This indicates strong bonding ability. The presence of free electron gas demonstrates the metallic bonding properties of Au and Mo atomic layers (fig. S2A). Furthermore, the electrostatic potential profile reveals that the carrier tunneling probability of the ALB interface is 100% ($\phi_t = 0$), whereas the barrier height at the vdW interface is 3.37 eV and the barrier width is 1.11 Å ($\phi_t > 0$) (right graph in Fig. 1, A and B). In addition, the local density of states indicates that the ALB region below the Au electrode is metallic, and the band bending in the ALB contact and MoS_2 channel heterojunction region is very steep in actual space (~0.5 nm), which allows for a high-probability tunneling of charge carriers in the space charge region in the on state (fig. S2B)—i.e., exhibiting good ohmic contact. This is similar to the situation in Si. We also observed multiple hybrid bands composed of Mo d orbitals and Au d orbitals intersecting with the Fermi level (E_F) (Fig. 1C) in ALB contact. By contrast, for the vdW contact, the bands intersecting with E_F are primarily composed of Au d orbitals, with the lowest point of the MoS_2 conduction band located above E_F (Fig. 1D), indicating that ALB contact greatly enhances band coupling. All of these indicate that ALB contact can achieve extremely low R_C. Moreover, the bonding energy of the ALB interface obtained through calculation reaches 0.281 eV/Å^2, which is a highly cohesive bonding interface and is 5.4 times as strong as that of the vdW interface (0.052 eV/Å^2) (Fig. 1E). This highly cohesive coherent bonding interface will guarantee the thermomechanical stability of contact interface and the entire contact structure (Fig. 1F). However,

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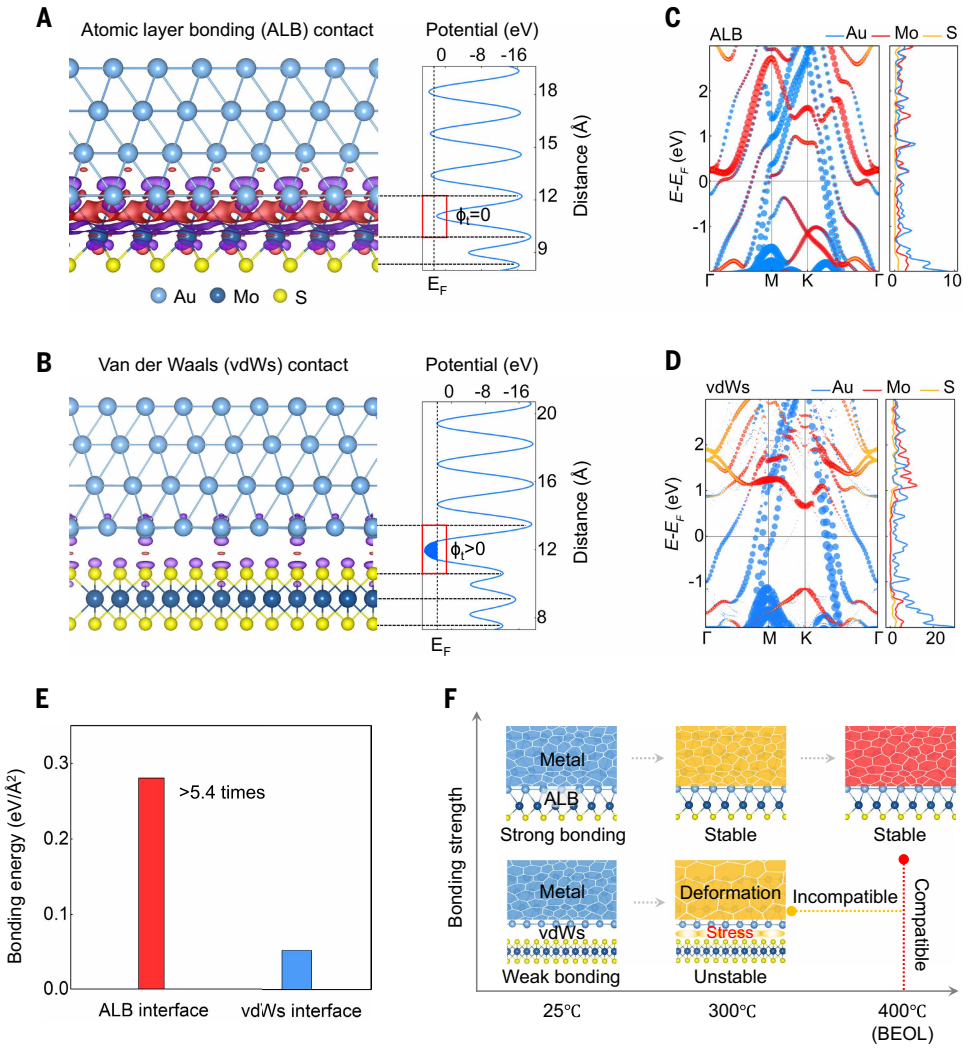


Fig. 1. Comparison of ALB contact and vdW contact. (**A** and **B**) Differential charge density of ALB contact (A) and vdW contact (B) of Au and MoS_2 system (red, positive; purple, negative). The isosurface level is 2.5×10^{-3} e bohr^{-3}. The graph on the right is the corresponding electrostatic potential profile along the vertical direction. (**C** and **D**) Electronic band structure and density of states of ALB contact (C) and vdW contact (D). (**E**) Bonding energy of ALB and vdW interface obtained from calculation. (**F**) Comparison of thermomechanical stability of ALB and vdW contacts before and after heating.

vdW contacts with weak bonding are susceptible to thermal deformation such as recrystallization of the interface electrode, leading to instability and stress concentration. We further demonstrated that the ALB contact system is structurally stable at temperatures up to 400°C by molecular dynamics calculations (fig. S2, C and D), meeting the thermal budget for back-end integration.

Realization of ALB contacts

In reality, ALB contact requires precise removal of the upper S atomic layer. The formation energy of S vacancies is smaller than that of Mo vacancies by a factor of 3 to 4, which provides a larger operating window for precise removal of S without damaging Mo (*27*–*29*). Notably, the transition-metal atomic layer lattice and the metal electrode lattice need to be highly matched; otherwise, large interface stress will be introduced. For the Au and MoS_2 system, Mo atoms can theoretically form a coherent bond with Au on the (111) plane by lattice relaxation, as shown in Fig. 1A. However, in reality, the lattice constant of Mo-Mo (a = 3.151 Å) is larger than that of Au-Au (b = 2.914 Å) on the (111) plane (Fig. 2, A and B), and the lattice mismatch is as high as 8%, making a stress-free and stable coherent bonding interface difficult to form. Ideal Au-Mo coherent bonding means Mo-Mo contraction or Au-Au moderate expansion.

Through DFT calculations, we found that when four layers of Au on the (111) plane are coherently bonded with -MoS, the Mo-Mo lattice shrinks by up to 6.22% (from 3.151 to 2.955 Å), whereas the expansion of Au-Au is only 1.39% (from 2.914 to 2.955 Å) (Fig. 2, C and D). This is because the contraction of Mo-Mo requires lower energy. From these results, the contraction of the Mo-Mo lattice to adapt to the metal lattice is the key to realizing coherent ALB contact in reality. In monolayer MoS_2, we observed that when the upper S atomic layer was completely removed, the Mo-Mo lattice shrank from the initial 3.151 to 2.968 Å because of the loss of upper support, with a shrinkage rate of 5.81% (Fig. 2E and fig. S3). This is very close to the 6.22% required for coherent bonding with Au, and the remaining mismatch can be completely adapted by lattice relaxation and elastic strain. This indicates that removal of the upper S atom provides both matching electronic and lattice structures. We also verified that the addition of more layers of Au does not have a substantial impact on the lattice matching of the bonding interface (fig. S4). The accumulation of Mo-Mo lattice shrinkage within a certain range will inevitably lead to the generation of local cracks (Fig. 2E). These cracks are partially filled with initially slowly deposited Au (red area in Fig. 2F). Moreover, these cracks can provide expansion space for lattice relaxation at the ALB interface when heated, depleting the internal stress of lattice relaxation while further increasing interface continuity, ensuring that the interface has both uniform electrical conductivity and high thermomechanical stability.

In practice, when we used ultrasoft argon (Ar) plasma to precisely etch away the upper S atomic layer of monolayer MoS_2, we obtained an etching interface that conforms to theoretical calculations (Fig. 2G). The finely controlled process is shown in fig. S5 and in the materials and methods. The boundary between the etched and unetched areas is very sharp, with no crack penetration, ensuring the construction of short-channel transistors (fig. S6A). The decrease in Mo3d binding energy of ~0.5 to ~0.6 eV obtained from x-ray photoelectron spectroscopy (XPS) at the etching interface further confirms the effective removal of top S (fig. S6B). Subsequently, we examined the ALB contact formed in multilayer MoS_2 and Au through high-angle annular dark-field scanning transmission electron microscope (HAADF-STEM). Multilayer MoS_2 is more conducive to clearly observing the atomic arrangement and comparing the lattice contraction. We observed that the Mo atoms form a nearly perfect coherent bond with Au (111) atoms (Fig. 2H). The Mo-Mo lattice (2.592 Å) at the bonding interface is almost identical to the Au-Au lattice (2.601 Å) and is smaller than that in bottom nonbonded MoS_2 (2.802 Å), confirming Mo-Mo lattice contraction at the bonding interface, consistent with the calculated results. The Mo-Mo lattice at the bonding interface shrinks by 7.49%

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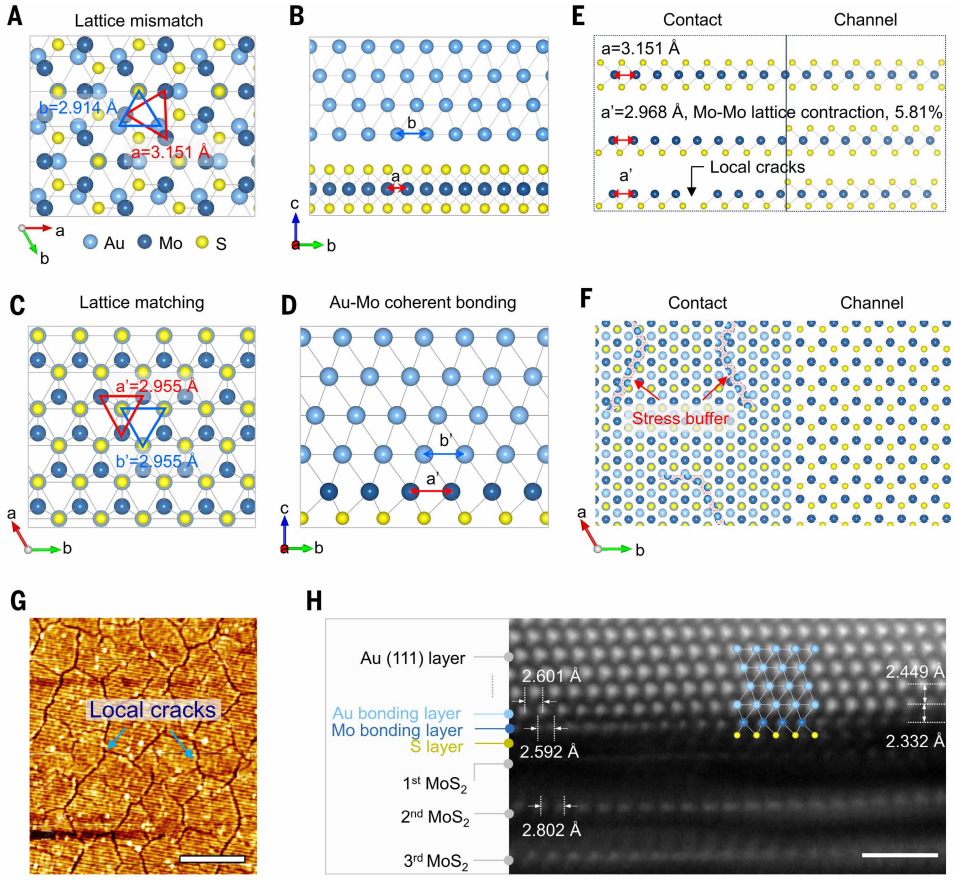


Fig. 2. Realization of ALB contacts between MoS_2 and Au electrode. (**A** and **B**) Bottom (A) and side views (B) of the optimized structure when Au (111) atoms contact MoS_2 monolayer. The Mo-Mo lattice constant (3.151 Å, red triangle side length) is larger than the Au-Au lattice constant (2.914 Å, blue triangle side length). (**C** and **D**) Bottom (C) and side (D) views of the coherent ALB contact formed by the contraction of the Mo-Mo lattice and the appropriate expansion of the Au-Au lattice. (**E**) Mo-Mo lattice contraction and local crack generation after removing the upper S atomic layer of MoS_2 calculated by DFT. (**F**) Overall bottom view of the ALB contact and channel. (**G**) AFM morphology of monolayer MoS_2 after the upper S atoms are etched away. Scale bar, 500 nm. The local cracks caused by the contraction of Mo-Mo lattice are clearly visible. (**H**) HAADF-STEM image of the ALB interface. Scale bar, 1 nm.

compared with the nonbonded part, which is greater than the 6.22% calculated by DFT. This is related to the lattice relaxation of the actual sample after heating. The interplanar spacing between Au and Mo atomic layers (2.332 Å) at the bonding interface is much smaller than that at the Au-MoS_2 vdW interface (4.225 Å) (fig. S7, A and B) and also smaller than the conventional vdW distance (*30*), indicating the formation of non-vdW bonding. The strong Au-Mo bonding even "sucks up" the adjacent MoS_2 layer (marked "1st MoS_2" in Fig. 2H), causing the vdW spacing to change. In addition, XPS results indicate that the binding energy of Mo at the ALB interface is ~0.3 to ~0.4 eV lower than when removing the upper S, confirming the effective Au-Mo bonding (fig. S7C). Overall, the ALB contact interface has a metallic chemical state, highly ordered coherent lattice, and very high bonding energy, ensuring extremely low R_C and excellent thermomechanical stability. By contrast, some reported cases of introducing certain vacancies in the contact area often exhibit localized electronic structures, severe lattice distortions, and low bonding energies, resulting in high R_C and poor stability (*31*, *32*).

Electronic properties of ALB contacts

Then, we evaluated the electronic properties of ALB contacts in monolayer MoS_2 field-effect transistors (FETs) (see materials and methods for fabrication and measurement details). Figure 3A is a schematic diagram of a MoS_2 transistor. Figure 3, B and C, shows the transfer characteristics of variable-channel MoS_2 transistors at V_{bias} of 0.1 V and R_C extracted by transfer-length method (TLM). In pristine conditions, the drain-source current (I_{ds}) increases uniformly as the channel length (L_{CH}) decreases. After annealing at 300°C, the I_{ds} exhibits a marked increase, attributed to the enhanced quality of the ALB. After annealing at 400°C, I_{ds} decreases slightly, which is mainly due to thermal damage of the unpackaged channel rather than degradation of the ALB interface (fig. S8). Accordingly, we obtained a R_C of 170 Ω·μm for the pristine FETs, which decreased to 70 and 110 Ω·μm after progressive annealing at 300° and 400°C. To our knowledge, this is the only type of contact in 2D semiconductors that maintains extremely low resistance after long-term high-temperature annealing. Figure S9 presents the transfer characteristics and R_C of several other groups of FETs. The pristine R_C is ~100 to ~200 Ω·μm, and after 300°C annealing, it is often <100 Ω·μm, with a minimum value of 61 Ω·μm. We determined the Schottky barrier height (SBH) of the pristine ALB contact to be 28 meV, because the pristine bonding interface was not perfect. After annealing at 300° and 400°C, SBH decreased to an extremely low or negligible value (fig. S10), consistent with theoretical calculations. We then examined the limiting performance of the short-channel FET. Figure 3D shows the transmission electron microscope (TEM) image of 30-nm short-channel MoS_2 FET, and Fig. 3, E and F, shows the corresponding transport and output performance of the same device after 300°C annealing. This FET exhibits a switching ratio of 10^7, a drain-induced barrier lowering (DIBL) of 30 mV/V, and a minimum subthreshold swing (SS) of 100 mV/dec. When the source-drain bias (V_{ds}) is only 0.7 V, I_{ds} reaches saturation, and the maximum I_{ds} reaches 1.1 mA/μm, representing the optimal current for the monolayer MoS_2 FET. Figure S11 presents the transport and output characteristics of three transistors with L_{CH} of ≤100 nm. All devices exhibited excellent electrical performance and outstanding thermal stability.

We further examined the transport properties of MoS_2 transistors with ALB contact formed with different S:Mo ratios. Consistent with the above conclusions, transistors with ALB contacts with S:Mo ratio close to 1:1 (almost all upper S is removed) have the highest on-off ratio, the largest on-current, and the best thermomechanical stability (fig. S12C). Insufficient or excessive etching at the bonding interface either fails to ensure the stability of the transistor or causes a decrease in the switching ratio and the on-current (fig. S12, A, B, and D). The durable thermomechanical stability of the ALB contact was further demonstrated after FET annealing at 300°C for 5 hours and 400°C for 3 hours (fig. S12, E and F). Our transistor exhibits the best heat resistance, surpassing all reported 2D transistors (fig. S12G). Conventional hard plasma etching and low-vacuum metal deposition also are

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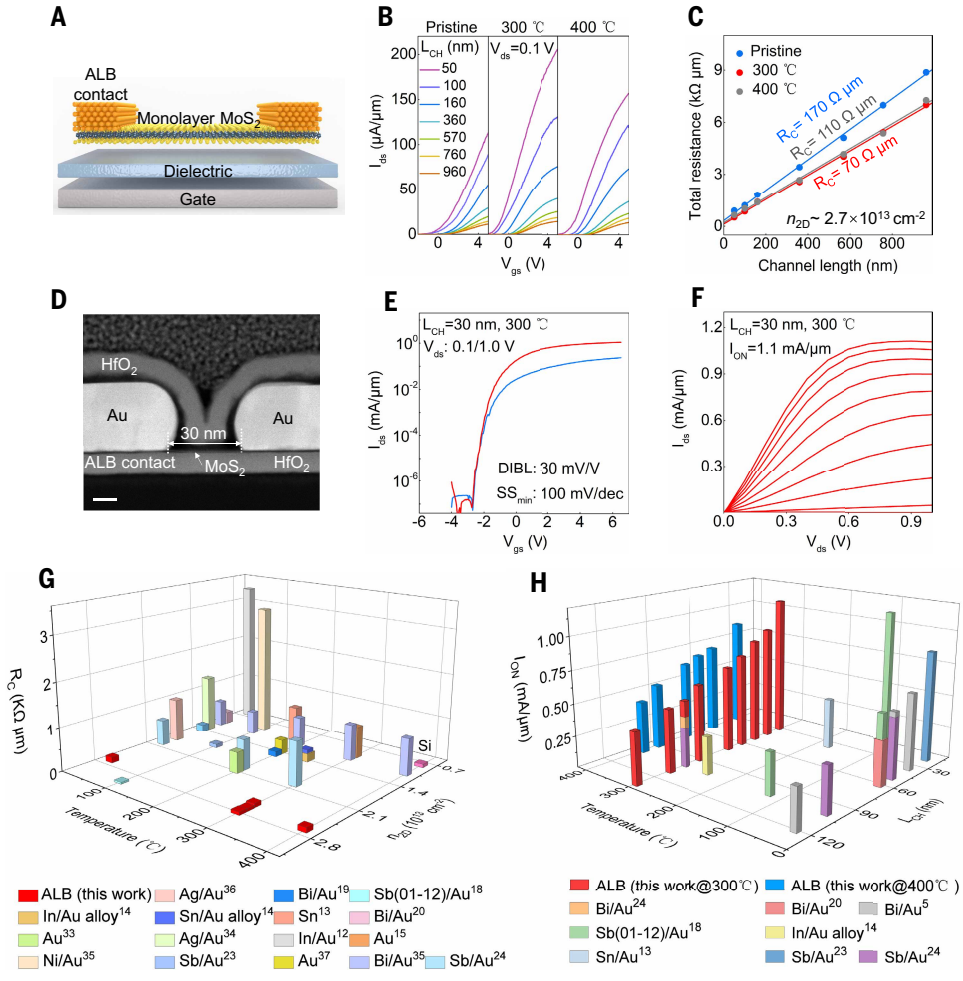


Fig. 3. Electronic properties of ALB contacts. (**A**) Diagram of MoS_2 transistor with ALB contact. (**B**) Transfer curves of a typical TLM structure with L_{CH} ranging from 50 to 960 nm at V_{bias} of 0.1 V. (**C**) R_C extracted by TLM method in (B). (**D**) Cross-sectional TEM image of a 30-nm short-channel MoS_2 FET. Scale bar, 10 nm. The top HfO_2 is deposited only when testing TEM and was not used to gate electrical transport. (**E**) Transfer curves of the same device after 300°C annealing in (D). (**F**) Output curves of the same device in (D). Gate-source voltage (V_{gs}) is from −3.5 to 6.5 V. (**G**) Comparison of R_C and thermomechanical stability of ALB contacts with other contacts. (**H**) Comparison of I_{ON} (V_{ds} = 0.7 V) and high-temperature resistance of the advanced short-channel monolayer MoS_2 transistor reported in the literature.

not conducive to achieving stable bonding contacts, and device performance is poor (fig. S12, H and I). In addition, by statistically comparing the performance of transistors with different channel widths, we have verified that the reliability of the final ALB contact interface is very high and will not be affected by potential cracks at the initial etched interface (fig. S13).

We next benchmarked our device performance with existing contact schemes. We first summarized the relationship between R_C, carrier concentration (n_{2D}), and reported withstood temperature for the state-of-the-art contact types for MoS_2 transistors in Fig. 3G. Our ALB contacts maintain the lowest R_C after high-temperature annealing at 300° and 400°C, surpassing all reported contact types (*13*–*15*, *18*–*22*, *24*, *33*–*37*). We then compared the I_{ON} (on state current density at V_{ds} = 0.7 V) and high-temperature resistance of short-channel monolayer MoS_2 transistors (Fig. 3H). Our devices can achieve high I_{ON} after annealing at 300° and 400°C, whereas most transistors with other types of contacts can only achieve room temperature performance (*13*, *14*, *18*–*21*, *23*, *24*). At V_{ds} of 0.7 V, the I_{ON} of our 30-nm short-channel MoS_2 transistor exceeds the high-performance target of the International Roadmap for Devices and Systems (IRDS) 2028 (*38*). Table S1 shows the detailed performance of monolayer MoS_2 transistors with different contacts.

Enhanced thermomechanical stability of the ALB contact

ALB interface brings outstanding thermomechanical stability. This advantage is also reflected in the effective resistance to recrystallization plastic deformation of metal electrodes at high temperatures, thereby improving the stability of the entire contact structure (Fig. 4A). By contrast, the bonding strength of the Au direct contact interface is insufficient, and the electrode is prone to recrystallization deformation when heated (Fig. 4B). At this time, the stress is easily transferred to contact and channel regions. The grain orientation spread map obtained by electron backscatter diffraction demonstrates that the degree of recrystallization of the Au on the ALB interface is substantially lower than that of Au on a MoS_2 interface under the same heating conditions (Fig. 4, C and D). As a result, the short-channel MoS_2 transistor with ALB contact has very stable contact structure before and after heating (Fig. 4E), whereas the recrystallization deformation of the direct Au contact causes the channel to stretch by 85% (from 110 to 204 nm) (Fig. 4F). We confirmed that the MoS_2 transistor with ALB contact is not affected by stress before and after heating (Fig. 4G), whereas the channel with Au direct contact suffers severe tensile stress (Fig. 4H) (*25*, *26*, *39*, *40*).

Subsequently, we compared the thermomechanical stability of different types of Au contact, including ALB contact, top contact, edge contact, and vdW contact (Fig. 4I). After annealing at 300°C, except for the devices with ALB contacts, the switching performance of transistors with other contacts dropped or was lost to varying degrees. This is due to the serious channel stress concentration or contact detachment (edge contact) caused by the recrystallization deformation of electrode. Furthermore, we have demonstrated in long-channel MoS_2 transistors (L_{CH} = 1 μm) that the thermal instability caused by direct contact between different types of metal electrodes is common, and ALB strategies can improve their thermal stability (fig. S14). However, high-performance and high-reliability short-channel transistors often require careful consideration of the lattice matching at the ALB interface and the oxidation resistance of the metal electrode; otherwise, thermal stability cannot be guaranteed (fig. S15).

We conducted statistical analysis using XPS on 20 points within a 2-inch (5.08 cm) wafer (*41*, *42*) range (with a test area of ~400 μm per point), verifying the excellent scalability and repeatability of the ultrasoft Ar plasma in removing S and forming ALB contact (fig. S16). Then, we tested the performance of the array transistor on a 2-inch wafer (fig. S17A). All devices exhibited excellent switching characteristics after annealing at 300° and 400°C, demonstrating the scalability of the ALB contact (fig. S17, B and C). We compared the normalized I_{ON} of MoS_2 transistors with ALB, Bi/Au, and Sb/Au contacts (fig. S17D).

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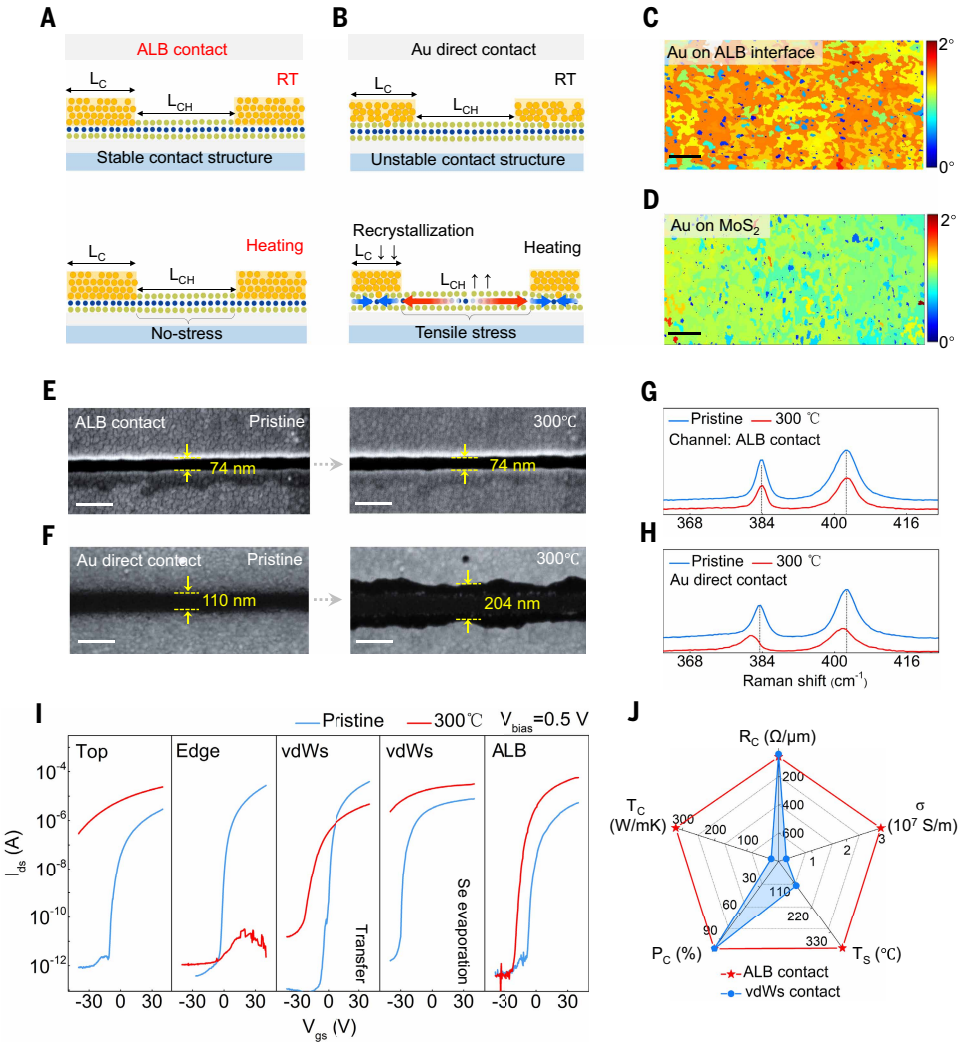


Fig. 4. Enhanced thermomechanical stability of the ALB contact. (**A**) Diagram of stable ALB contact before and after heating. L_C, contact length; RT, room temperature. (**B**) Diagram of stress concentration caused by recrystallization deformation of unstable Au direct contact after heating. (**C** and **D**) Grains orientation spread map of deposited Au electrodes on ALB interface (C) and on MoS_2 interface (D). Scale bar, 1 μm. The 0° to 2° scale represents the recrystallization grain orientation. The larger the orientation, the lower the degree of recrystallization deformation. (**E** and **F**) Scanning electron microscopy images of short-channel MoS_2 transistors with ALB contact (E) and Au direct contact (F) before and after heating. Scale bar, 200 nm. (**G** and **H**) Raman spectrum of channel area of MoS_2 transistor with ALB contact (G) and Au direct contact (H) before and after heating. (**I**) Transport properties of MoS_2 transistor with ALB contact and different types of Au contacts before and after annealing. (**J**) Radar plot of our ALB contact versus Sb vdW contact, benchmarked against industrial-grade contact performance. P_C, process compatibility; R_C, contact resistance (Ω·μm); σ, electrical conductivity (S/m); T_C, thermal conductivity (W/mK); T_S, thermomechanical stability (°C).

After annealing at 300°C, the I_{ON} of ALB devices increase by 400% compared with the pristine current. Even after annealing at 400°C, it remains 235% higher than the pristine current. By contrast, the performance of the Bi/Au and Sb/Au contacted device decayed substantially after annealing at 300°C, and Bi/Au failed completely at 400°C (*24*). These fully validate that ALB contacts play an irreplaceable role in improving stability. In addition, we have confirmed through calculations and experiments that the ALB contact achieved by compensating etching on the basis of bond energy differences (*43*) can be widely extended to other TMD semiconductors, such as tungsten disulfide and molybdenum diselenide (fig. S18 and materials and methods).

ALB contacts not only have ultralow R_C and excellent thermomechanical stability but also have much better heat-dissipation performance than mainstream vdW contacts. Taking the ALB contact formed by Au and MoS_2 as an example, both Au and Mo exhibit excellent thermal conductivity (κ_{Au}: 317 W/mK; κ_{Mo}: 138 W/mK) and electrical conductivity (σ_{Au}: 4.10×10^7 S/m; σ_{Mo}: 1.87×10^7 S/m) that far exceed semimetals such as Sb (κ_{Sb}: 24 W/mK; σ_{Sb}: 2.76×10^6 S/m) (fig. S1, B and C). Therefore, the heat-dissipation characteristics of ALB contact interface are bound to far exceed those of semimetal vdW contact. In this way, we can confirm that ALB contact comprehensively surpasses vdW contact, is comparable to covalent bond contact, and can meet industrial requirements (Fig. 4J).

Conclusions

We proposed and demonstrated ALB contact between 2D TMD semiconductor and metal by establishing metallic coherent bonding interface. ALB contact exhibits ultrastrong band coupling and high cohesive bonding strength and achieves notable synergistic breakthrough in ultralow R_C and thermomechanical stability, surpassing all reported contact configurations and comparable to covalent bonding contacts in semiconductor industry. Thanks to the ALB contact, the R_C of monolayer MoS_2 transistor is as low as 70 Ω·μm, the contact thermomechanical stability is as high as 400°C, and the on-state current after high-temperature annealing is 1.1 mA/μm. All of these surpass equivalent silicon technologies and meet the high-performance target of the 2028 IRDS Roadmap. We demonstrated the scalability of ALB contacts and their generalizability to different 2D TMDs and metals. The ALB contact overcomes the inherent problem of vdW semiconductors, greatly increases the industrial compatibility of vdW electronic devices, and is expected to lead the transformation of 2D electronics from "lab to fab."

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SUPPLEMENTARY MATERIALS

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